

Electrodeposited self-assembled molecules for perovskite photovoltaics

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Preventing the detachment of self-assembled molecules (SAMs) and enhancing their passivation effect on perovskites are critical challenges for improving the performance and stability of perovskite solar cells^{1–3}. Electrodeposited SAMs provide a route to improve coverage uniformity and anchoring robustness on conductive substrates beyond the limitations of conventional solution processing. Here we use potential-cycled electrodeposition to promote molecular rearrangement and re-anchoring of SAMs, resulting in a uniform and dense layer on an indium tin oxide substrate with enhanced anchoring capability. Building on this base SAM, functional units are grown via electrochemical oxidative coupling to form tailored coupled carbazole phosphonic SAMs, yielding power conversion efficiencies of 26.8% for laboratory-scale solar cells and 21.3% for solar modules (65 cm²).

Self-assembled molecules (SAMs) are increasingly favoured as hole transport layers (HTLs) in perovskite solar cells (PSCs) owing to ultrathin, tightly packed interfaces and efficient charge extraction^{4,5}. Although PSCs incorporating SAMs have achieved power conversion efficiencies (PCEs) approaching 27% and have maintained over 90% of the initial performance after 1,000 h (refs. 6–9), there is still plenty of room to further improve the functionality of SAMs for enhancing the efficiency and stability of PSCs. Typically, solution-processed SAMs face diffusion and coverage uniformity challenges, which hinder the fabrication of PSC modules and their real-world deployment¹⁰. A critical challenge for long-term device stability is SAM desorption and diffusion during operation^{11,12}. Given the limited density of –OH groups on indium tin oxide (ITO), solution-based methods, such as immersion, spin-coating or blade-coating, typically produce a molecular layer that comprises both chemically anchored and physisorbed fractions, thereby limiting anchoring strength and molecular layer homogeneity.

Electrodeposition enables controlled interfacial formation and provides a route to address these challenges^{13–16}. By applying a controlled electrochemical potential, weak Brønsted acids undergo deprotonation, coupled with hydrogen evolution and interact with metal surfaces^{17,18}. In our electrodeposition route, phosphonic-acid deprotonation is integrated with potential-controlled interfacial formation, molecular rearrangement and re-anchoring, resulting in a dense and compact layer with an increased chemically anchored fraction^{19–21}. Industrial electroplating tooling can be leveraged to enable the electrodeposition of SAMs over large areas, facilitating the scalable fabrication of perovskite solar modules with reliable reproducibility²².

Beyond forming a robust base SAM, electrodeposition offers a versatile strategy for functionalizing SAMs by introducing further functional units via oxidative coupling. This approach provides a modular

platform for the rational incorporation of diverse functional groups, enabling precise regulation of perovskite film formation and interfacial passivation^{23–25}. Using this strategy, based on the electrodeposited base SAM, we further incorporated functional groups such as cyano (–CN), pyridyl (–Py) and adamantyl (–Ad). The functionalized SAMs show different interactions with the perovskite layer, resulting in a champion PCE of 26.8% (certified at 26.4%) for inverted PSCs, retaining 96% of the initial efficiency over 2,500 h under 1 sun illumination at 65 °C.

Anchoring stability of SAMs on ITO

The anchoring of spin-coated SAMs is strongly affected by the surface M–OH density of the ITO²⁶, and the inherent roughness of the substrate causes (2-(9H-carbazol-9-yl)ethyl)phosphonic acid (2PACz) molecules to accumulate in surface concavities (Fig. 1a)^{27,28}. By comparison, electrodeposition couples –PO₃H₂ deprotonation with in situ self-repair during potential cycling, where molecular rearrangement and re-anchoring progressively fill undercovered regions on ITO, thereby improving interfacial anchoring and film uniformity (Fig. 1b and Supplementary Fig. 1).

Indium tin oxide substrates, which were used as the working electrodes during electrodeposition, were immersed in a 2PACz electrolyte solution (Supplementary Figs. 2–4). Under reductive potential, the –PO₃H₂ headgroup of 2PACz undergoes deprotonation to generate –PO₃H[–] and –PO₃^{2–} species (Supplementary Figs. 5–7 and Supplementary Table 1). Compared with –PO₃H₂ headgroups, which are more dependent on surface M–OH groups for dehydration–condensation anchoring, these deprotonated species provide a less hydroxyl-dependent pathway for forming P–O–M interactions with accessible In and Sn surface sites on ITO. In this way, accessible metal sites complement surface M–OH sites on the heterogeneous ITO surface, increasing the

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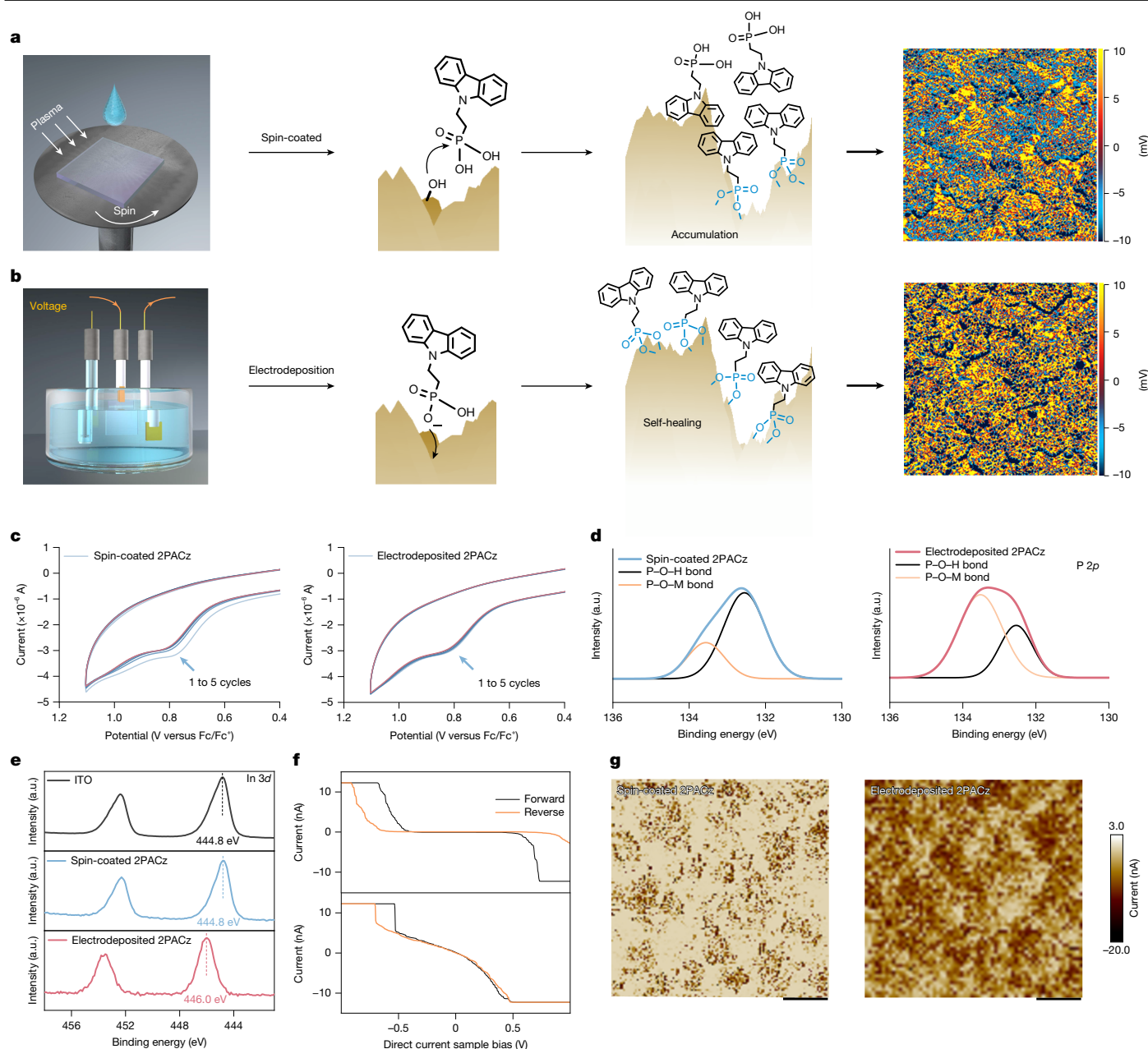


Fig. 1 | Deposition-route dependence of 2PACz. **a, b**, Schematics of spin-coated (**a**) and electrodeposited (**b**) 2PACz, illustrating the dominant interfacial scenarios rather than mutually exclusive anchoring pathways, with IR-AFM signatures indicating more uniform coverage for the electrodeposited sample. **c**, Cyclic voltammograms of ITO substrates bearing spin-coated and electrodeposited 2PACz in blank electrolyte, used to probe the electrochemical stability of surface-bound 2PACz. **d**, High-resolution P 2p XPS spectra of ITO substrates bearing spin-coated and electrodeposited 2PACz, with deconvoluted P-O-H and P-O-M components. Peak assignment was guided by reference

number of effective anchoring sites and facilitating the formation of a dense SAM layer.

To compare the dominant anchoring pathways under spin-coating and electrodeposition, we synthesized an ¹⁸O-labelled analogue (¹⁸O-2PACz) (Supplementary Fig. 8 and Supplementary Scheme 1) and performed TOF-SIMS isotope tracing. For ¹⁸O-2PACz, M-OH-assisted dehydrative condensation would favour P-¹⁶O-M species through incorporation of substrate-derived oxygen, whereas oxygen-retaining binding of deprotonated phosphonate species would preserve ¹⁸O in P-¹⁸O-M species (Supplementary Fig. 9). Spin-coated ¹⁸O-2PACz showed

spectra: P-O-H from powder 2PACz and P-O-M from depth-profiled electrodeposited 2PACz in the region closest to the ITO surface, consistent with P-O-M anchoring. **e**, In 3d XPS spectra of ITO-2PACz prepared by spin-coating and electrodeposition. **f**, Local I-V curves measured by c-AFM on ITO-2PACz prepared by spin-coating (top) and electrodeposition (bottom). **g**, c-AFM current maps of ITO-SAM samples prepared by spin-coating (left) and electrodeposition (right), acquired at -1.0 V bias, mapping nanoscale current-density distributions. Scale bars, 400 nm.

a dominant ¹⁶O-2PACz-related ion envelope (*m/z* 273–275), consistent with a larger contribution from dehydrative condensation with surface M-¹⁶OH. By contrast, electrodeposited ¹⁸O-2PACz retained a stronger ¹⁸O-2PACz-related ion envelope (*m/z* 277–279), supporting a larger contribution from oxygen-retaining phosphonate binding under the applied potential (Supplementary Fig. 10).

To assess interfacial stability, spin-coated and electrodeposited 2PACz films were immersed in a blank electrolyte, and their cyclic voltammetry curves were recorded (Fig. 1c). Over multiple cyclic voltammetry scans, the carbazole peak current at 0.8 V for spin-coated

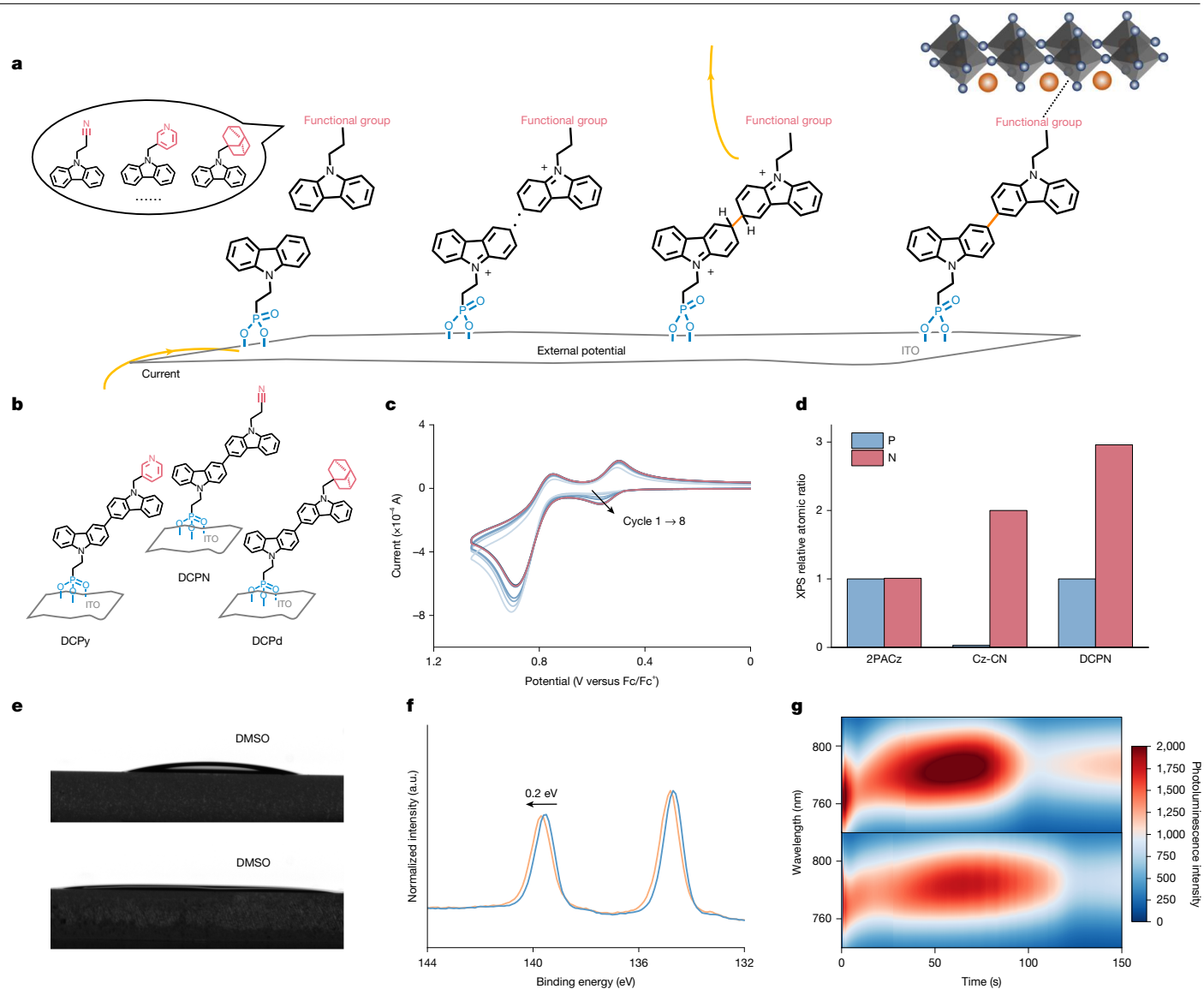


Fig. 2 | Modular functionalization of electrodeposited SAMs by oxidative coupling. **a**, Schematic of the electrochemical oxidative coupling route to functionalized SAMs. **b**, Representative products of modular SAM functionalization: DCPN (-CN), DCPy (-Py) and DCPd (-Ad). **c**, Cyclic voltammograms recorded during electrochemical oxidative coupling to form DCPN-functionalized SAMs. **d**, Relative P and N atomic ratios of 2PACz, Cz-CN

and DCPN from XPS. **e**, Dimethyl sulfoxide (DMSO) contact angles on electrodeposited 2PACz (top) and DCPN (bottom). **f**, Pb 4f XPS spectra of perovskite films on electrodeposited 2PACz (blue) and DCPy (orange). **g**, In situ photoluminescence monitoring during perovskite film formation on electrodeposited 2PACz (top) and DCPd (bottom).

2PACz decreased steadily, consistent with the desorption of physisorbed 2PACz into the electrolyte (Supplementary Fig. 11), whereas electrodeposited 2PACz had negligible desorption. Moreover, depth-profiled X-ray photoelectron spectroscopy (XPS) revealed that, towards the ITO surface, the P 2*p* envelope shifted to a higher binding energy, consistent with the enrichment of P–O–M near the interface (Supplementary Fig. 12). By comparison, XPS of the electrodeposited 2PACz film revealed a higher P–O–M fraction (Fig. 1d), which explains the suppressed desorption and improved electrochemical stability. The positive binding-energy shift of the In 3*d* peak further corroborates this conclusion, consistent with a deposition-route-dependent difference in interfacial bonding (Fig. 1e and Supplementary Figs. 13 and 14). Furthermore, density functional theory (DFT) simulations indicate that during the electrodeposition process, the PO_3^{2-} group interacts with metal sites on the ITO surface, resulting in stronger bonds and shorter bond lengths, which indicate enhanced anchoring strength (Supplementary Fig. 15).

Furthermore, electrodeposited 2PACz had more complete and uniform coverage on ITO, as revealed by infrared atomic force microscopy (IR-AFM). Given that the SAM is an ultrathin molecular layer (~1 nm) relative to the ITO roughness (~5 nm), and that the spin-coating defines the morphology in a single-step process without re-equilibration, 2PACz consequently accumulates in concavities, yielding heterogeneous coverage. Conversely, electrodeposited 2PACz under potential cycling progressively fills uncoated regions on ITO, delivering uniform coverage through in situ self-repair (Supplementary Figs. 16–18).

Conductive AFM (c-AFM) *I*–*V* measurements revealed differences in charge transport between spin-coated and electrodeposited 2PACz, and the electrodeposited 2PACz had reduced current hysteresis (Fig. 1f). Furthermore, c-AFM mapping of spin-coated 2PACz revealed distinct discrete current spots. Regions of low current were observed at recessed ITO sites, which aligns with the inhomogeneous surface coverage of the 2PACz layer (Supplementary Fig. 19). By contrast, electrodeposited 2PACz yielded a homogeneous current distribution, owing to its

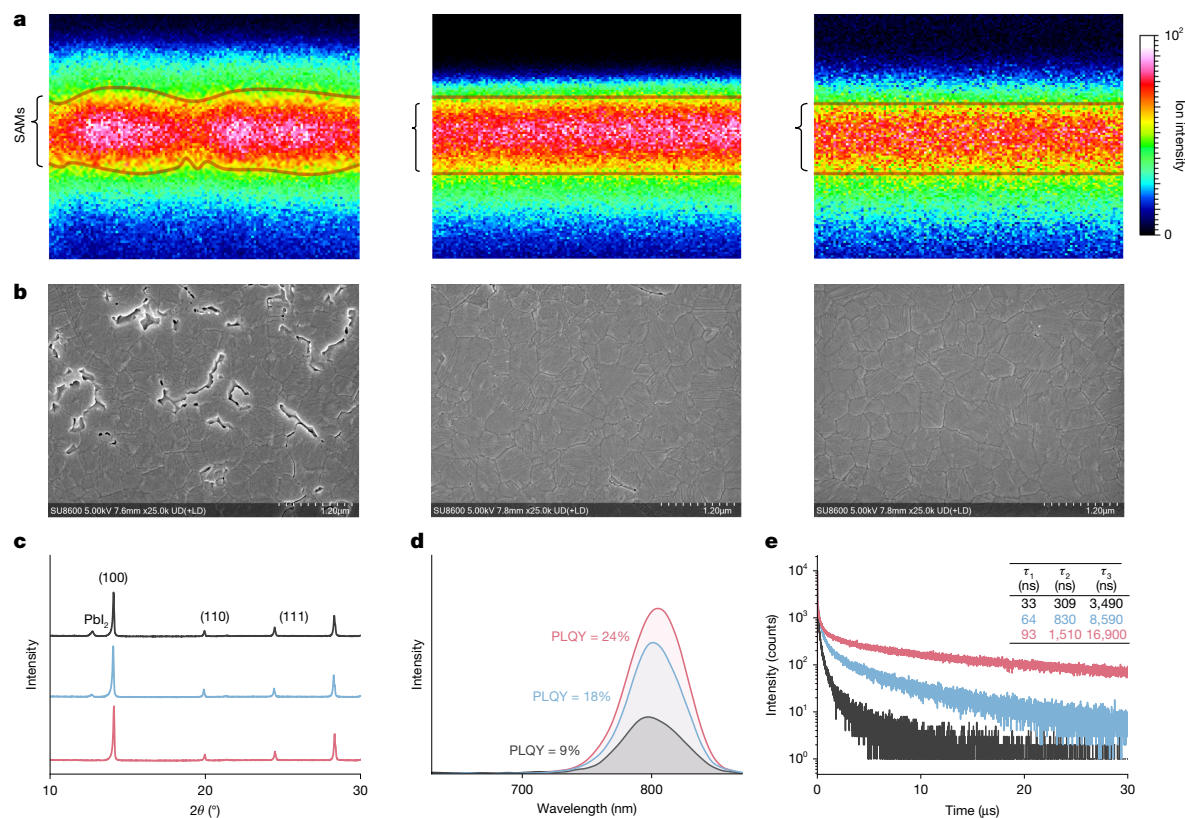


Fig. 3 | Effect of SAM on the morphology and optical properties of perovskite. **a**, TOF-SIMS depth profiles of perovskite films on spin-coated 2PACz (left), and electrodeposited 2PACz (middle) and DCPN (right), highlighting a more uniform vertical distribution for electrodeposited SAMs. **b**, SEM images of the buried perovskite bottom interface on spin-coated 2PACz, electrodeposited 2PACz and DCPN. **c**, XRD patterns of perovskite films on

spin-coated 2PACz (black), and electrodeposited 2PACz (blue) and DCPN (red). **d**, Photoluminescence quantum yield (PLQY) of the perovskite film on spin-coated 2PACz (black), and electrodeposited 2PACz (blue) and DCPN (red). **e**, TRPL decays of perovskite films on spin-coated 2PACz (black), and electrodeposited 2PACz (blue) and DCPN (red).

uniform morphology (Fig. 1g). Notably, the electrodeposited SAM approach was scalable to large-area substrates ($10 \times 10 \text{ cm}^2$), yielding a uniform and dense molecular layer while preserving SAM integrity (Supplementary Figs. 20–22).

Functionalized electrodeposited SAMs

Building on the electrodeposited 2PACz base layer, an oxidative potential was applied to graft Cz–R onto surface-facing sites (R = functional groups) while preserving the integrity of the underlying 2PACz molecular layer (Fig. 2a, and Supplementary Figs. 23 and 24). In this modular design, functional groups can be introduced to match the requirements of the perovskite layer, thereby customizing SAM properties to regulate perovskite microstructure and optoelectronic properties^{29–31}. Taking cyano (–CN), pyridyl (–Py) and adamantyl (–Ad) groups as examples, we denote the resulting functionalized coupled carbazole-phosphonic SAMs as DCPN, DCPy and DCPd, respectively (Fig. 2b). These SAMs enhance perovskite wetting, improve perovskite-layer passivation and regulate perovskite nucleation, respectively.

Taking electrodeposited DCPN as an example, the –CN group was introduced via oxidative coupling between the electrodeposited 2PACz base layer and *N*-(2-cyanoethyl)carbazole (Cz–CN) in the electrolyte (Fig. 2c). Under oxidative potential, a redox feature at 0.6 V emerged from the second cycle, consistent with carbazole dimer formation^{32–34}, and supported by ultraviolet–visible absorption and Raman spectra (Supplementary Figs. 25–27). Effective coupling was further corroborated by time-of-flight secondary ion mass spectrometry (TOF-SIMS), which showed that the $\text{C}_{12}\text{NH}_9\text{C}_2\text{H}_4\text{CN}^+$ fragment ion is assignable to the

fused-ring segment of the DCPN molecule (Supplementary Fig. 28). X-ray photoelectron spectroscopy elemental analysis further revealed an N:P ratio of 2.96 for the carbazole dimer film, close to the stoichiometric value expected for the DCPN molecule (Fig. 2d, and Supplementary Figs. 29 and 30), indicating efficient on-surface coupling.

Subsequently, –Py and –Ad functional groups were grafted via the same electrodeposition process (Supplementary Fig. 31), and the dependence of interfacial properties on functional group identity was investigated. For electrodeposited DCPN, contact-angle measurements and DFT simulations indicated that the –CN group enhances the surface wettability toward the perovskite precursor solution (Fig. 2e, and Supplementary Figs. 32 and 33)³⁵. For electrodeposited DCPy, perovskite films on electrodeposited DCPy showed a higher XPS Pb 4f binding energy (Fig. 2f), which is consistent with stronger passivation by the –Py group. For electrodeposited DCPd, in situ photoluminescence monitoring revealed that the perovskite films on electrodeposited DCPd showed a lengthened induction period before the growth regime (Fig. 2g), indicating that the –Ad group modulates crystallization kinetics and retards crystal growth^{36–39}. Collectively, electrodeposition provides a modular interfacial platform that enables the rational introduction of functional groups to regulate perovskite film formation and interfacial passivation.

Perovskite film quality

To clarify the influence on perovskite film morphology and optoelectronic quality, DCPN was taken as a representative case. TOF-SIMS cross-section analysis of the *x*–*z* plane showed that spin-coated 2PACz

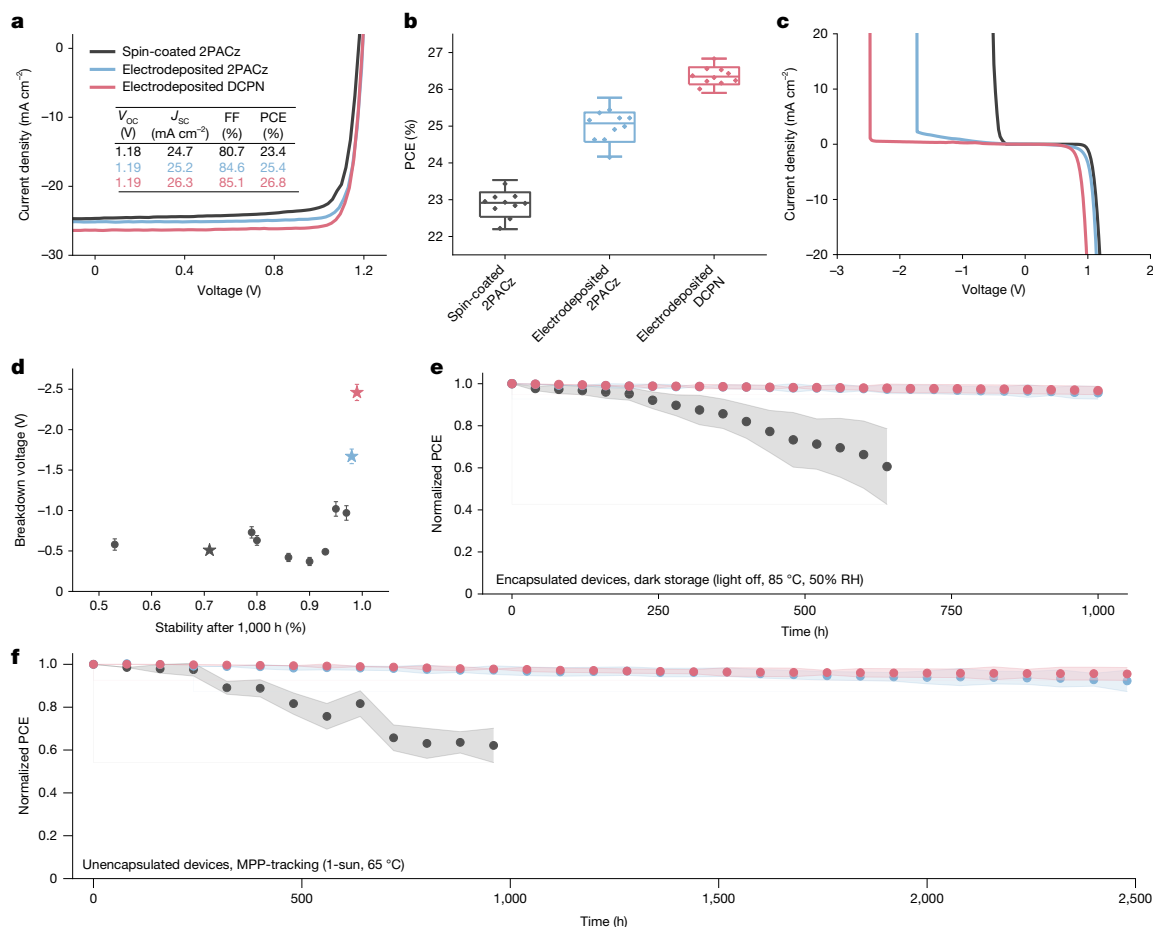


Fig. 4 | SAM-dependent device performance and stability. **a**, $J-V$ curves (reverse scan) and photovoltaic parameters for devices with different SAMs. FF, fill factor. **b**, PCE statistics for devices with different SAMs ($n = 10$). **c**, Dark $J-V$ curves for devices with different SAMs, highlighting reverse-bias behaviour. **d**, Statistical summary of operational stability and reverse-bias breakdown voltage (V_{rb} ; mean $\pm$ s.d.; $n = 5$) for PSCs with different SAMs. **e**, Dark-storage

stability (ISOS-D; light off). Devices were aged at 85 °C and 50% relative humidity (RH) in the dark, and PCE retention was measured periodically (mean $\pm$ s.d.; $n = 5$). **f**, Operational stability (ISOS-L; continuous maximum power point (MPP) tracking). Devices were tracked under 1-sun-equivalent illumination at 65 °C in a nitrogen-filled glovebox (mean $\pm$ s.d.; $n = 5$).

had pronounced aggregation (Fig. 3a)^{40–42}, yielding a non-uniform interface; scanning electron microscopy (SEM) further revealed visible voids at the buried perovskite bottom layer (Fig. 3b), features suggestive of compromised film quality and increased defect density^{43,44}.

By contrast, electrodeposited 2PACz and DCPN had more homogeneous coverage, yielding compact perovskite grains with narrow grain boundaries and diminished intergranular voids. Moreover, SEM images of the perovskite top surface revealed that films grown on electrodeposited 2PACz and DCPN showed more uniform grains, with the –CN functional group in DCPN further promoting crystallization, leading to larger grains (Supplementary Fig. 34). X-ray diffraction (XRD) corroborated enhanced crystallinity in the DCPN-based perovskite film, as evidenced by a weaker lead iodide (PbI_2) reflection and a stronger (100) texture (Fig. 3c). Furthermore, TOF-SIMS revealed that DCPN suppressed PbI_2 accumulation at the top interface of perovskite films, thereby alleviating interfacial defects (Supplementary Fig. 35). Moreover, compared with spin-coated Cz–CN, which shows upward redistribution, the anchoring of DCPN via the $-\text{PO}_3\text{H}_2$ group effectively prevents the upward migration of the functional group (Supplementary Fig. 36). Consistent with these interfacial improvements, ultraviolet photoelectron spectroscopy measurements indicated a more favourable energetic alignment at the DCPN-modified perovskite buried interface, characterized by a reduced valence-band-related offset for hole extraction and an enlarged conduction-band-related offset for electron blocking (Supplementary Fig. 37).

To correlate morphological improvements with optoelectronic properties, we measured the photoluminescence quantum yield (PLQY) and time-resolved photoluminescence (TRPL). DCPN-based perovskite films exhibited the highest PLQY of 24% (Fig. 3d), indicating reduced trap-assisted recombination and improved defect passivation. Time-resolved photoluminescence further showed an extended lifetime of 16,900 ns (Fig. 3e), consistent with suppressed non-radiative recombination. Differential-lifetime analysis of the TRPL decays revealed faster hole transfer and longer carrier lifetime for DCPN, consistent with reduced first-order interfacial recombination (Supplementary Fig. 38). Together, these results indicate that DCPN improves perovskite buried-interface optoelectronic quality by suppressing non-radiative recombination and accelerating interfacial charge transfer.

Device performance

To further assess the photovoltaic performance of PSCs, inverted devices with a structure of $\text{ITO}/\text{SAMs}/\text{Cs}_{0.05}(\text{FA}_{0.95}\text{MA}_{0.05})_{0.95}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3/\text{C}_{60}/\text{bathocuproine (BCP)}/\text{Ag}$ were fabricated⁴⁵. As shown by the current density–voltage ($J-V$) curves (Fig. 4a,b, Supplementary Fig. 39 and Supplementary Table 2), the champion device based on spin-coated 2PACz achieved a power conversion efficiency (PCE) of 23.4%, whereas the champion device based on electrodeposited 2PACz reached 25.4% after optimizing the electrodeposition parameters

(Supplementary Figs. 40–42). Subsequently, functionalized electrodeposited SAMs exerted varying effects on device performance (Supplementary Fig. 43), with DCPN delivering the highest efficiency, achieving a PCE of 26.8% with an open-circuit voltage (V_{oc}) of 1.19 V, a short-circuit current density (J_{sc}) of 26.3 mA cm⁻² and a fill factor of 85.1%. The external-quantum-efficiency-integrated J_{sc} was 25.2 mA cm⁻², in agreement with the J - V value (Supplementary Fig. 44). The champion PCE of 26.8% was further certified at 26.4% by an accredited third party (Supplementary Fig. 45), highlighting the effectiveness of functionalized electrodeposited SAMs in enhancing device efficiency.

Shockley–Queisser–referenced-loss analysis and a pseudo- J - V overlay were used to identify the origin of improvements in photovoltaic parameters. DCPN-based devices had a higher fill factor and J_{sc} , which were consistent with improved interfacial uniformity and wettability, facilitating improved interfacial carrier transport and more efficient collection (Supplementary Figs. 46 and 47; Supplementary Tables 3 and 4). Concurrently, the absolute PLQY of the device increased from 4% for spin-coated 2PACz to 8% for DCPN (Supplementary Fig. 48), indicating suppressed non-radiative recombination and consistent with the higher V_{oc} . Device-physics measurements further supported improved interfacial charge extraction and suppressed recombination in DCPN-based devices. The apparent built-in potential extracted from Mott–Schottky analysis increased from 0.81 V for spin-coated 2PACz to 0.88 V for electrodeposited DCPN (Supplementary Fig. 49), while trap-state distribution analysis showed a reduced trap-state response for the DCPN-based device (Supplementary Fig. 50). Under illumination, electrochemical impedance spectroscopy showed a lower interfacial charge-transfer resistance and a higher recombination resistance (Supplementary Fig. 51), and the ideality factor extracted from light-intensity-dependent V_{oc} decreased from 1.36 to 1.23 (Supplementary Fig. 52). Transient photocurrent and transient photovoltage measurements gave a shorter extraction time of 0.64 μ s and a longer carrier lifetime of 15.6 μ s, respectively (Supplementary Fig. 53). Collectively, the introduction of DCPN improved charge extraction and reduced recombination in device performance.

Furthermore, reverse-bias benchmarking revealed early breakdown at -0.5 V for spin-coated 2PACz devices, consistent with leakage through locally weak or incompletely covered interfacial regions and subsequent localized degradation^{46,47}. In contrast, devices incorporating electrodeposited DCPN sustained reverse-bias breakdown voltages (V_{fb}) of approximately -2.5 V (Fig. 4c). To contextualize these findings, we compiled stability metrics for reported SAM-based PSCs and measured their reverse-bias breakdown voltages (Fig. 4d, Supplementary Fig. 54 and Supplementary Table 5). Beyond reverse-bias tolerance, device stability was evaluated under dark-storage, operational and open-circuit ageing conditions. Under dark-storage ageing at 85 °C and 50% relative humidity with light off, the spin-coated 2PACz devices retained 61% of their initial PCE after 640 h, whereas the electrodeposited DCPN devices retained 97% after 1,000 h (Fig. 4e). Under continuous maximum power point tracking at 65 °C under 1-sun-equivalent illumination in a nitrogen-filled glovebox, devices incorporating electrodeposited 2PACz and DCPN retained 92% and 96% of the initial PCE after 2,500 h, respectively (Fig. 4f, and Supplementary Figs. 55 and 56). Accelerated open-circuit ageing under 1-sun-equivalent illumination at 65 °C showed the same trend, with the DCPN-based devices retaining 95% of the initial PCE after 700 h (Supplementary Fig. 57). Together, these results show that electrodeposited SAMs support improved reverse-bias tolerance and operational stability, supporting their potential for practical PSC applications.

We further evaluated the electrodeposited SAM strategy in module fabrication. Perovskite solar modules with an active area of 65 cm² achieved a champion efficiency of 21.3% (certified at 20.9%) (Supplementary Figs. 58 and 59, and Supplementary Table 6)⁴⁸. The electrodeposited molecular layer also retained its functionality under realistic storage and transportation environments (Supplementary

Fig. 60), demonstrating the feasibility of a pre-electrodeposited SAM supply chain, where electrodeposition can be performed industrially.

Conclusion

In summary, we used electrodeposition to fabricate functionalized SAMs with improved anchoring and coverage uniformity on ITO. Under potential cycling, 2PACz underwent molecular rearrangement and re-anchoring, forming a more uniform molecular layer. A second functional unit was further introduced through electrochemical oxidative coupling to regulate the buried perovskite interface. These interfacial changes improved charge extraction and reduced recombination, resulting in an efficiency of 26.8% for small-area perovskite solar cells and 21.3% for 65 cm² solar modules, together with improved device stability under light, heat, humidity and reverse-bias stress. This approach links potential-controlled SAM formation with modular functionalization and supports the use of electrodeposited molecular contacts in perovskite photovoltaics.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10844-6>.

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Methods

Materials

Except for tetrabutylammonium hexafluorophosphate (TBAPF₆), all reagents were used as received without further purification. We sourced 2PACz, Cz–CN, *N*-(2-pyridyl)carbazole (Cz–Py) and *N*-(2-adamantylethyl)carbazole (Cz–Ad) from Dongguan Volt-Amp Optoelectronics. Formamidinium iodide, PbI₂, caesium iodide, methylammonium bromide, methylamine hydrochloride and piperazine hydroiodide were purchased from Xi'an Yuri Solar. *N,N*-Dimethylformamide (DMF), DMSO, TBAPF₆, ethanol, acetonitrile, chlorobenzene and isopropanol were anhydrous and purchased from J&K Scientific.

Fabrication of spin-coated 2PACz

The spin-coated 2PACz film fabrication began by cleaning ITO glass substrates via ultrasonic treatments in isopropanol, laboratory detergent, deionized water and isopropanol (in sequence), followed by drying at 80 °C. Next, the cleaned substrates underwent a 5 min oxygen plasma treatment before being transferred into a glove box. A 2PACz solution (1 mg ml⁻¹ in ethanol) was then spin-coated onto the ITO at 3,000 r.p.m. for 30 s, with an acceleration of 1,500 r.p.m. s⁻¹, before being thermally annealed at 100 °C for 10 min.

Fabrication of electrodeposited 2PACz

Before electrolyte preparation, TBAPF₆ was recrystallized from ethanol three times. After purification, ITO glass substrates were ultrasonically cleaned in isopropanol, laboratory detergent, deionized water and isopropanol (in sequence) and dried at 80 °C. The electrolyte was then prepared by dissolving 2PACz (1 mg ml⁻¹) and TBAPF₆ (0.1 M) in a 1:10 mixture of acetonitrile and ethanol. All subsequent electrochemical steps were performed inside a glove box: the cleaned ITO was immersed in the electrolyte and connected as the working electrode in a three-electrode cell with a platinum (99.99%) counter electrode and a RE-7 (Ag/Ag⁺) reference electrode for the non-aqueous environment. Potentials were scanned from 0.4 to -2.15 V (referenced to Fc/Fc⁺) at 50 mV s⁻¹ for five cycles on the electrochemical workstation. During this scan, 2PACz on ITO underwent reductive interfacial formation at the working-electrode–electrolyte interface, yielding an electrodeposited 2PACz molecular layer on the ITO working electrode. Immediately after deposition, the films were rinsed sequentially with ethanol and acetonitrile to remove residual electrolyte and free 2PACz, and then annealed at 100 °C for 5 min.

Fabrication of electrodeposited DCPN

Building on the electrodeposited 2PACz film, an electrochemical oxidative-coupling step was used to introduce the second functional unit and form the DCPN molecular layer. A solution of Cz–CN (1 mg ml⁻¹) and TBAPF₆ (0.1 M) in acetonitrile served as the electrolyte. The electrodeposited ITO–2PACz substrate was immersed in this electrolyte and connected as the working electrode in a three-electrode configuration, with a platinum (99.99%) plate as the counter electrode and a RE-7 (Ag/Ag⁺) reference electrode for non-aqueous environments. A potential sweep from 0 to 1.05 V (referenced to Fc/Fc⁺) was applied via the electrochemical workstation, at 50 mV s⁻¹ for eight cycles. After deposition, rinsing with acetonitrile effectively removed the self-coupled Cz–CN species and residual electrolyte without affecting the anchored DCPN, then annealed at 100 °C for 5 min.

DCPy and DCPd were prepared using the same procedure, except that Cz–CN was replaced with Cz–Py and Cz–Ad, respectively.

Perovskite solar-cell fabrication

Small-area cell fabrication. For the preparation of Cs_{0.05}(FA_{0.95}MA_{0.05})_{0.95}Pb(I_{0.95}Br_{0.05})₃, a 1.5 M perovskite precursor solution was prepared by mixing caesium iodide, FAI, PbI₂ and methylammonium

bromide in a DMF:DMSO = 4:1 (vol/vol) solvent mixture according to the stoichiometric ratio. The solution was stirred for 12 h, with a further 3 mol% PbI₂ and 10 mol% methylamine hydrochloride added to improve crystallization. Based on the previously prepared SAMs, we then used these various ITO–SAMs substrates to proceed with the subsequent steps of perovskite device fabrication. The perovskite layer was prepared by spin-coating the perovskite solution at 5,000 r.p.m. (ramping up from 1,000 r.p.m. over 5 s) for 30 s, during which 200 µl of chlorobenzene was dropped onto the spinning substrate in the final 12 s. The resulting film was then annealed at 110 °C for 30 min to form the perovskite layer. Next, a piperazine hydroiodide solution (0.3 mg ml⁻¹ in 2-propanol) was spin-coated at 5,000 r.p.m. for 30 s and subsequently annealed at 100 °C for 5 min to serve as an interfacial passivation layer⁴⁵. A 20 nm C₆₀ layer was deposited by thermal evaporation under high vacuum to function as the electron transport layer, followed by a 6 nm BCP cathode buffer. Finally, an Ag electrode (100 nm) was evaporated through a shadow mask, yielding devices with an active area of 0.04 cm².

Module fabrication. Modules were fabricated on ITO substrates bearing the SAM prepared above. Before SAM formation, P1 laser scribing (40 W, 30 µm) was performed on bare ITO. The FA_{0.9}Cs_{0.1}PbI₃ ink was prepared at 1.55 M in DMF:*N*-methyl-2-pyrrolidone (NMP) = 4:1 (v/v), stirred for 12 h. Perovskite films were blade coated at a stage speed of 4 mm s⁻¹. Immediately after coating, a vacuum flash was applied for 10 s at 5 Pa, followed by annealing at 100 °C for 40 min. Phenethylammonium iodide (0.5 mg ml⁻¹ in isopropanol) was then blade coated as an interfacial passivation⁴⁸. The electron-transport contact was formed by thermal evaporation of C₆₀ (20 nm), followed by low-temperature atomic-layer deposition of SnO₂ (20 nm). After C₆₀ and SnO₂ deposition, P2 laser scribing (35 W, 70 µm) was conducted. Silver (100 nm) was then thermally evaporated to form the top electrode and busbars, followed by P3 laser scribing (33 W, 30 µm). The module active area was 65 cm².

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting the findings of this study are available in the main text or Supplementary Information. Further raw characterization data generated during this study are available from the corresponding authors on reasonable request. Source data are provided with this paper.

Code availability

No custom code was used to generate the central conclusions of this study. The analysis procedures are described in the Methods and Supplementary Information.

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Author contributions L.Y. and N.L. conceived the project and supervised the study. Z.X., B.W., G.L. and Y.W. developed and performed the electrodeposition of 2PACz and the electrochemical oxidative coupling used to form DCPN. F.T. and S.Y. fabricated the perovskite solar cells and conducted the stability measurements. Z.X. and Y.L. synthesized the isotope-labelled materials and prepared the isotope-tracing samples. C.P. performed the TOF-SIMS measurements. X.L. performed the DFT calculations. S.T., J.W. and X.K. conducted the TRPL and PLQY measurements. Z.Y., Y.L., M.L., Y.L., Y.S. and W.M. conducted the electrochemical impedance spectroscopy, transient photocurrent and transient photovoltage measurements. Z.X., X.L. and F.T. drafted the manuscript. N.L. and L.Y. substantively revised the manuscript. All authors commented on the manuscript.

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Competing interests L.Y. and Z.X. are inventors on a Chinese patent application related to this work, filed by South China University of Technology with the China National Intellectual

Property Administration (application no. CN202511221491.7; publication no. CN121250496A). The other authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	Small-area cells had an active area of 0.04 cm ² . Perovskite solar modules had an active area of 65 cm ² . Explain why this information is not reported/not relevant.
Method used to determine the device area	☒ Yes ☐ No	Small-cell areas were defined by black metal aperture masks. Module active areas were defined by the laser-scribed module layout. Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	Forward and reverse J-V scans were provided over voltage windows extending from -0.1 V to slightly above the Voc of each device group and back to -0.1V.
Voltage scan conditions	☒ Yes ☐ No	Scan rate:0.2V/s Step size:0.02V Delay:50 ms Explain why this information is not reported/not relevant.
Test environment	☒ Yes ☐ No	J-V measurements were performed in a nitrogen-filled glovebox under simulated AM 1.5G illumination. Explain why this information is not reported/not relevant.
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	Provide a description of the protocol. No preconditioning was used in this work.
Stability of the J-V characteristic	☒ Yes ☐ No	Device stability was evaluated under dark-storage, continuous MPP-tracking and open-circuit ageing conditions. Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes ☐ No	Minor hysteresis was observed for devices herein. Explain why this information is not reported/not relevant.
Related experimental data	☒ Yes ☐ No	J-V curves under reverse and forward scans were provided. Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	EQE curves were provided. Explain why this information is not reported/not relevant.
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The integrated Jsc values from EQE were consistent with the Jsc values from J-V measurements. Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Provide a description of the measurement conditions. Not applicable; single -junction inverted perovskite architecture.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Using a Keithley 2400 sourcemeter under 1 sun irradiation from an AM 1.5G solar simulator (SS-F5, Enlitech). The illumination intensity was calibrated to 100 mW/cm ² using a certified Si reference cell. Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The Si reference cell used to calibrate the solar simulator was calibrated and certified. It was used to set the AM 1.5G illumination intensity to 100 mW/cm ² before J–V measurements. Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☐ Yes ☒ No	Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements. Spectral mismatch was not explicitly calculated. The solar simulator was calibrated using a certified Si reference cell, and EQE-integrated J _{sc} values were compared with J–V-derived J _{sc} values to verify consistency.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	A black metal aperture mask of 0.04 cm ² was used for small-area cells. The module active area was 65 cm ² . Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area. Not measured. All small-area devices were measured using the same 0.04 cm ² black metal aperture mask to avoid area-dependent variation.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	The champion small-area device was certified by the South China National Center of Metrology, accredited to ISO/IEC 17025:2017. TÜVNORD certification report for a 65 cm ² PSC module with electrodeposited SAMs, tested in accordance with IEC 61215-2:2021. Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	Supplementary Figures 45 and 59. Explain why this information is not reported/not relevant.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	For J–V performance statistics, 10 cells were tested for each device group. For stability measurements, 5 cells were tested for each condition unless otherwise stated. Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Device performance statistics are shown as distributions for 10 cells per group. Stability data shown as distributions for 5 cells per group. Stability data are reported as mean ± s.d. from five devices per condition. Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Long-term stability was evaluated under three conditions. Dark-storage ageing was performed at 85 °C and 50% relative humidity with light off. Operational stability was measured by continuous MPP tracking at 65 °C under one-sun-equivalent illumination in a nitrogen-filled glovebox. Accelerated open-circuit ageing was also performed at 65 °C under one-sun-equivalent illumination. Device temperature was monitored using a thermocouple attached to a dummy device placed adjacent to the tested cells. Explain why this information is not reported/not relevant.

